Plume and surface feature structure and compositional effects on Europa's global exosphere

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The article presents a detailed exploration of how a potential plume source on Europa could influence its global exosphere, which may exhibit intricate spatial structures and temporal variations in both density and composition. Researchers have developed a Monte Carlo exosphere model that incorporates a water plume source, enriched with various organic and nitrile species, to analyze these interactions.

In their modeling approach, the team considered several factors affecting the exosphere, including Europa's gravity, which plays a crucial role in returning plume ejecta back to the moon's surface. They also examined how thermal desorption and re-sputtering processes spread adsorbed and exospheric materials across Europa's surface. The study takes into account various mechanisms, such as sputtering, radiolysis, surface adsorption, regolith diffusion, polar cold trapping, and the re-sputtering of adsorbed materials, in order to assess the spatial distribution and temporal evolution of the exospheric density and composition.

Four distinct case scenarios were analyzed:

- **Case A**: An equatorial flyby through an exosphere generated solely by sputtering, with no plume contributions.

- **Case B**: A flyby over a localized region marked by sputtered terrain, specifically a 'macula' enriched with non-ice species.

- **Case C**: A south polar plume scenario, reminiscent of Enceladus, during an equatorial flyby.

- **Case D**: A direct flyby through a south polar plume, allowing for direct analysis of plume constituents.

The findings from these models serve as a predictive basis for future telescopic observations, including those from the Hubble Space Telescope (HST) and the James Webb Space Telescope (JWST), as well as planned missions to the Jovian system by both NASA and the European Space Agency (ESA). The researchers also applied spacecraft trajectories to their model to anticipate the potential exospheric compositions that could be encountered during proposed flybys of Europa. This analysis aims to clarify the spatial and temporal relationships between spacecraft measurements and the surface and plume source compositions.

Given the forthcoming ESA JUICE mission and NASA's Europa multiple-flyby mission, scheduled to reach Europa by the end of the next decade, comprehending the origin, structure, dynamics, and composition of its exosphere is of paramount importance. These insights will be critical for understanding Europa's surface and sub-surface composition and chemistry.

The article emphasizes that Europa's known exospheric components include oxygen (O and O_2), hydrogen (H and H_2), and water (H_2O), originating from surface water ice through radiolytic processes. Trace elements such as sulfur dioxide (SO_2), sodium (Na), chlorine (Cl), and potassium (K) have also been detected, likely stemming from radiolysis reactions and the dissociation of surface salts. The research acknowledges the extensive observational and modeling efforts that have been dedicated to studying Europa's exosphere since its discovery over two decades ago.

Collectively, these studies represent significant advancements in understanding the complex interactions within Europa's exosphere and underscore the urgency of further investigation as new missions are launched.

Cassini Ion and Neutral Mass Spectrometer: Enceladus Plume Composition and Structure

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The report presents a comprehensive analysis of data collected by the Cassini spacecraft's Ion and Neutral Mass Spectrometer (INMS) during its closest approach to Enceladus on July 14, 2005, at 19:55:22 UTC. At a proximity of just 168.2 kilometers above the moon's surface, Cassini encountered a significant atmospheric plume and coma, which were detectable in the INMS data extending over 4000 kilometers from Enceladus.

The analysis reveals that the composition of the atmospheric plume is predominantly made up of water vapor (H_2O), which constitutes approximately 91% of the detected material, along with 3.2% carbon dioxide (CO_2). Additionally, the data indicated the presence of a species with a mass-to-charge ratio of 28 daltons, which is likely to be either carbon monoxide (CO) or molecular nitrogen (N_2). Methane (CH_4) was also detected at a level of about 1.6%.

Trace components in the plume include small quantities of acetylene (C_2H_2) and propane (C_3H_8), both present in concentrations less than 1%. Ammonia (NH_3) was noted at very low levels, not exceeding 0.5%. These findings suggest that the primary source of the plume is concentrated near the south polar region of Enceladus, as inferred from the radial and angular distributions of gas density measured during the flyby. This is consistent with independent evidence indicating that there is a significant localized contribution to the plume, distinct from a more uniform global source detected during the outbound leg of the flyby.

The INMS instrument was uniquely positioned, pointing within 60 degrees of the direction of the spacecraft's motion and traveling at a relative velocity of approximately 8 km/s. This configuration allowed for unprecedented measurements of the plume's gas composition and spatial distribution. Unlike previous flybys that had the INMS oriented in the anti-ram direction, this approach enabled the detection of neutral gases linked to Enceladus.

The INMS itself is a dual-source quadrupole mass spectrometer that covers mass-to-charge ranges from 0.5 to 8.5 daltons and from 11.5 to 99.5 daltons. It integrates classic closed and open ionization source configurations to measure both inert and reactive species. For the primary data reported, the closed source was utilized, which allows neutral gases to flow into a spherical antechamber that thermally equilibrates with its walls before passing to an electron ionization source. Ionization occurs via electron impact at 70 eV, enhancing sensitivity, although the 60-degree orientation limited some measurements.

Data acquisition involved mass scans every 4.6 seconds when the spacecraft was within 500 kilometers of Enceladus. The collected spectra were combined to improve the signal-to-noise ratio, with separate background corrections applied for ingress and egress data to account for variations before and after the flyby.

Mass peaks corresponding to the primary atmospheric constituents were identified at 18 daltons (H_2O), 44 daltons (CO_2), 28 daltons (N_2 or CO), and 16 daltons (CH_4). Minor species such as C_2H_2 and C_3H_8 were also detected, while potential trace levels of NH_3 and hydrogen cyanide (HCN) were considered based on their presence in the data. Calibration of the instrument responses for various constituents was performed prior to flight operations, ensuring the analysis' reliability.

The derived atmospheric composition emphasizes the significance of local sources in shaping the plume, contributing to our understanding of Enceladus as a potentially habitable environment and the dynamics of its eruptions. The findings highlight the importance of continued exploration and study of Enceladus's intriguing features and the broader implications for planetary science.